

**Robert D. McLaughlin**

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**PSEG**

**NUCLEAR**

10 CFR 50.73

LR-N24-0054

September 5, 2024

U.S. Nuclear Regulatory Commission  
ATTN: Document Control Desk  
Washington, DC 20555-0001

Hope Creek Generating Station  
Renewed Facility Operating License No. NPF-57  
Docket No. 50-354

Subject: Licensee Event Report 2024-001-01, Invalid Primary Containment  
Integrated Leak Rate As-Found Test (ILRT) Supplement

PSEG Nuclear LLC submits licensee event report (LER) 2024-001-01 per  
10 CFR 50.73(a)(2)(ii)(A) and 10 CFR 50.73(a)(2)(v)(C).

There are no regulatory commitments contained in this letter.

Please contact Ms. Tammy Morin at (856) 339 -2286 if you have questions.

Sincerely,

Robert D. McLaughlin  
Plant Manager, Hope Creek Generating Station

Attachment: Licensee Event Report

cc: USNRC Regional Administrator – Region 1  
USNRC NRR Project Manager – Hope Creek  
USNRC Senior Resident Inspector – Hope Creek  
NJ Department of Environmental Protection, Bureau of Nuclear Engineering

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to [Infocollections.Resource@nrc.gov](mailto:Infocollections.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

|                                                   |                                         |                           |                   |
|---------------------------------------------------|-----------------------------------------|---------------------------|-------------------|
| 1. Facility Name<br>Hope Creek Generating Station | <input checked="" type="checkbox"/> 050 | 2. Docket Number<br>00354 | 3. Page<br>1 OF 2 |
|                                                   | <input type="checkbox"/> 052            |                           |                   |

|                                                                                   |
|-----------------------------------------------------------------------------------|
| 4. Title<br>Invalid Primary Containment Integrated Leak Rate As-Found Test (ILRT) |
|-----------------------------------------------------------------------------------|

| 5. Event Date |     |      | 6. LER Number |                   |              | 7. Report Date |     |      | 8. Other Facilities Involved |                              |               |
|---------------|-----|------|---------------|-------------------|--------------|----------------|-----|------|------------------------------|------------------------------|---------------|
| Month         | Day | Year | Year          | Sequential Number | Revision No. | Month          | Day | Year | Facility Name                | <input type="checkbox"/> 050 | Docket Number |
| 05            | 03  | 2024 | 2024          | - 001 -           | 01           | 09             | 05  | 2024 | Facility Name                | <input type="checkbox"/> 052 | Docket Number |

|                        |                        |
|------------------------|------------------------|
| 9. Operating Mode<br>4 | 10. Power Level<br>000 |
|------------------------|------------------------|

**11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)**

|                                                    |                                                    |                                                    |                                                        |                                                    |                                     |
|----------------------------------------------------|----------------------------------------------------|----------------------------------------------------|--------------------------------------------------------|----------------------------------------------------|-------------------------------------|
| <input checked="" type="checkbox"/> 10 CFR Part 20 | <input type="checkbox"/> 20.2203(a)(2)(vi)         | <input checked="" type="checkbox"/> 10 CFR Part 50 | <input checked="" type="checkbox"/> 50.73(a)(2)(ii)(A) | <input type="checkbox"/> 50.73(a)(2)(viii)(A)      | <input type="checkbox"/> 73.1200(a) |
| <input type="checkbox"/> 20.2201(b)                | <input type="checkbox"/> 20.2203(a)(3)(i)          | <input type="checkbox"/> 50.36(c)(1)(i)(A)         | <input type="checkbox"/> 50.73(a)(2)(ii)(B)            | <input type="checkbox"/> 50.73(a)(2)(viii)(B)      | <input type="checkbox"/> 73.1200(b) |
| <input type="checkbox"/> 20.2201(d)                | <input type="checkbox"/> 20.2203(a)(3)(ii)         | <input type="checkbox"/> 50.36(c)(1)(ii)(A)        | <input type="checkbox"/> 50.73(a)(2)(iii)              | <input type="checkbox"/> 50.73(a)(2)(ix)(A)        | <input type="checkbox"/> 73.1200(c) |
| <input type="checkbox"/> 20.2203(a)(1)             | <input type="checkbox"/> 20.2203(a)(4)             | <input type="checkbox"/> 50.36(c)(2)               | <input type="checkbox"/> 50.73(a)(2)(iv)(A)            | <input type="checkbox"/> 50.73(a)(2)(x)            | <input type="checkbox"/> 73.1200(d) |
| <input type="checkbox"/> 20.2203(a)(2)(i)          | <input checked="" type="checkbox"/> 10 CFR Part 21 | <input type="checkbox"/> 50.46(a)(3)(ii)           | <input type="checkbox"/> 50.73(a)(2)(v)(A)             | <input checked="" type="checkbox"/> 10 CFR Part 73 | <input type="checkbox"/> 73.1200(e) |
| <input type="checkbox"/> 20.2203(a)(2)(ii)         | <input type="checkbox"/> 21.2(c)                   | <input type="checkbox"/> 50.69(g)                  | <input type="checkbox"/> 50.73(a)(2)(v)(B)             | <input type="checkbox"/> 73.77(a)(1)               | <input type="checkbox"/> 73.1200(f) |
| <input type="checkbox"/> 20.2203(a)(2)(iii)        |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(A)         | <input checked="" type="checkbox"/> 50.73(a)(2)(v)(C)  | <input type="checkbox"/> 73.77(a)(2)(i)            | <input type="checkbox"/> 73.1200(g) |
| <input type="checkbox"/> 20.2203(a)(2)(iv)         |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(B)         | <input type="checkbox"/> 50.73(a)(2)(v)(D)             | <input type="checkbox"/> 73.77(a)(2)(ii)           | <input type="checkbox"/> 73.1200(h) |
| <input type="checkbox"/> 20.2203(a)(2)(v)          |                                                    | <input type="checkbox"/> 50.73(a)(2)(i)(C)         | <input type="checkbox"/> 50.73(a)(2)(vii)              |                                                    |                                     |

☐ OTHER (Specify here, in abstract, or NRC 366A).**12. Licensee Contact for this LER**

|                                 |                                                    |
|---------------------------------|----------------------------------------------------|
| Licensee Contact<br>Tammy Morin | Phone Number (Include area code)<br>(856)-339-2286 |
|---------------------------------|----------------------------------------------------|

**13. Complete One Line for each Component Failure Described in this Report**

| Cause | System | Component | Manufacturer | Reportable to IRIS | Cause | System | Component | Manufacturer | Reportable to IRIS |
|-------|--------|-----------|--------------|--------------------|-------|--------|-----------|--------------|--------------------|
| D     | NH     | SEAL      | GE           | N                  |       |        |           |              |                    |

|                                        |                                                                              |  |  |  |                              |  |  |       |     |      |
|----------------------------------------|------------------------------------------------------------------------------|--|--|--|------------------------------|--|--|-------|-----|------|
| 14. Supplemental Report Expected       |                                                                              |  |  |  | 15. Expected Submission Date |  |  | Month | Day | Year |
| <input checked="" type="checkbox"/> No | <input type="checkbox"/> Yes (If yes, complete 15. Expected Submission Date) |  |  |  |                              |  |  |       |     |      |

**16. Abstract** (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On May 3, 2024, during a refueling outage, Hope Creek Generating Station (HCGS) performed Primary Containment Integrated Leak Rate Testing (ILRT) Type A test as required by Technical Specification 4.6.1.2. The as-found test was not completed successfully because the stabilization criteria was not met, and the test was invalidated. The as-left test was completed successfully prior to plant start-up.

This licensee event report (LER) supplement provides the results of a Root Cause Evaluation (RCE). The Direct Cause was inadequate bolt seating torque that led to loss of O-ring seal compression. The Root Cause was identified as inadequate procedural guidance for validating and measuring flange to drywell airlock surface gap.

This event was reported to the NRC as an eight-hour, non-emergency per 10 CFR 50.72(b)(3)(ii)(A) in Event Notification number 57103 on May 3, 2024. Subsequently, LER 2024-001-00 was submitted on July 2, 2024, reporting the event per 10 CFR 50.73(a)(2)(ii)(A). This LER supplement also identifies additional reporting criterion, 10 CFR 50.73(a)(2)(v)(C).

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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|                                                              |                                                |                                      |                      |                                 |                      |
|--------------------------------------------------------------|------------------------------------------------|--------------------------------------|----------------------|---------------------------------|----------------------|
| <b>1. FACILITY NAME</b><br><br>Hope Creek Generating Station | <input checked="" type="checkbox"/> <b>050</b> | <b>2. DOCKET NUMBER</b><br><br>00354 | <b>3. LER NUMBER</b> |                                 |                      |
|                                                              | <input type="checkbox"/> <b>052</b>            |                                      | <b>YEAR</b><br>2024  | <b>SEQUENTIAL NUMBER</b><br>001 | <b>REV NO.</b><br>01 |

**NARRATIVE****PLANT AND SYSTEM IDENTIFICATION**

General Electric - Boiling Water Reactor (BWR/4)  
System - Primary Containment

**IDENTIFICATION OF OCCURRENCE**

Discovery Date: May 3, 2024

**CONDITIONS PRIOR TO OCCURRENCE**

OPCON 4, Cold Shutdown. No other structures, systems, or components that could have contributed to the event were inoperable.

**DESCRIPTION OF OCCURRENCE**

On May 2, 2024, at 10:00 am, the Primary Containment Integrated Leak Test (ILRT) commenced. At 1:08 pm, the stabilization period started. At 6:30 pm there was evidence of leakage and walkdowns were initiated to determine the location of any leaks. At 8:30 pm a leak was found at the drywell airlock hatch to primary containment flange. At 11:00 pm, in accordance with procedures, an attempt to quantify the leak was made.

On May 3, 2024, at 3:05 am, a second attempt to quantify the leak was made and at 4:11 am, it was determined that a step change in the leakage rate had invalidated the as-found test.

On May 4, 2024, at 10:00 am, the Type A as-left ILRT was completed with satisfactory results.

**CAUSE OF EVENT**

The Direct Cause of the event was identified as inadequate bolt seating torque. This led to loss of O-ring seal compression, which created a leakage path through the air-lock flange. The RCE identified one Root Cause and one Contributing Cause.

The Root Cause was inadequate procedural guidance for validating and measuring flange to drywell airlock surface metal to metal contact. The procedure also did not provide steps to validate the torque values after rigging is relaxed or removed.

A Contributing Cause was identified which led to the invalid test. Individuals involved in the test did not fully assess the risk, consequences, or potential leakage adjustment, of installing a procedurally allowed pressure block on a Type B penetration without knowing the condition of the component.

**SAFETY CONSEQUENCES AND IMPLICATIONS**

The event occurred when HCGS was in OPCON 4, Cold Shutdown. The test was performed in an operating mode where the safety function of the primary containment was not required, and the as-left test was performed successfully prior to entering a mode when the primary containment is required. This is reportable as a degraded condition of a principal safety barrier because the stabilization criteria were not met.

An engineering analysis was performed. The analysis demonstrates the dose for the control room, exclusion area boundary, and low population zone were within regulatory dose limits under all conditions. Therefore, there were no safety consequences to site personnel or the public.

**CORRECTIVE ACTIONS**

A satisfactory Type A as-left ILRT was performed to verify Primary Containment integrity exists. The procedure for re-installation of the drywell has been revised to ensure metal to metal surface contact after installation of the airlock and an as-left LLRT on the joint will be performed. The Primary Containment ILRT procedure has been revised to require the program owner and test director to approve the installation of pressure blocks to Type B penetrations.

**COMMITMENTS**

There are no regulatory commitments in this LER.